RAG Assignment: Baseline vs. Re-Ranking

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GitHub: `https://github.com/CReativeRebe1/RAG_LMA_Assignment1`
Code: `rag_baseline_vs_rerank.ipynb` (all code and outputs; every LLM answer is saved in `results/`)

Setup

The document is India's Digital Personal Data Protection Act, 2023 (21 pages). I split it along its own structure (section, sub-section, clause) into 105 chunks of at most 200 tokens, and rebuilt the penalty table (the Schedule) row by row so each fine stays with its rule. Bi-encoder: `all-MiniLM-L6-v2` with a FAISS inner-product index.

Cross-encoder: `cross-encoder/ms-marco-MiniLM-L-6-v2`. LLM: `openai/gpt-oss-120b` on Groq, temperature 0. Both pipelines use the same prompt, which allows answering only from the 3 given chunks and otherwise replying that the extracts do not answer the question. I tested 32 plain-language questions and marked, for each, the chunk that really answers it.

1. Architecture comparison

A **bi-encoder** encodes the question and each chunk separately. Token outputs are mean-pooled into one 384-dimensional vector and normalised, so relevance is the cosine similarity of two vectors. The chunk vectors are computed once in advance; a query costs one encoding plus a vector search. Because each chunk is reduced to one fixed vector without knowing the question, a rule and its exception, or a "shall" and a "shall not", can end up very close, and a chunk that only repeats the question's words can rank high.

A **cross-encoder** reads both together as one input (`[CLS] question [SEP] chunk [SEP]`), with self-attention across all tokens of both texts, and outputs one score for the pair. It can see whether the chunk actually answers this specific question, but its scores cannot be stored in advance: every (question, chunk) pair needs a full forward pass at query time.

2. Rank shifts and answer comparison

Case	Question (short)	Correct chunk: baseline → re-ranked	Answer 1 (baseline)	Answer 2 (re-ranked)
Negation / role limit (Q09)	Must the Data Protection Officer be in India?	#5 → #1	"Extracts do not answer"	Yes, the DPO must be based in India (s.10(2))
Keyword trap / synonym (Q13)	Can I take back my permission?	#4 → #1	Yes, but built from Consent-Manager and deletion chunks; missed "at any time, as easily as given"	Full rule from s.6(4)–(5)
Negation / constraint (Q21)	Can a company refuse service if I won't share data it doesn't need?	#4 → #2	"Extracts do not answer"	Consent must be limited to the data needed (s.6(1))
Keyword trap / synonym (Q25)	App unused for years, must my data be deleted?	#5 → #3	Guessed "yes" from the general deletion rule; missed s.8(8)	Refused, even though s.8(8) was in its context
Penalty elsewhere (Q01)	Max fine for weak security?	#3 → #1	Rs 250 crore (correct)	Rs 250 crore (correct)
Regression (Q12)	Can I pretend to be someone else?	#2 → #7	No, you must not impersonate (s.15, correct)	Refused, the chunk was pushed out
Regression (Q20)	What decides the size of a fine?	#1 → #1	All 7 factors of s.33(2)	Only 5 factors; the chunk with (f)–(g) was dropped
Control (Q17)	Days to appeal a Board order?	#1 → #1	60 days (correct)	60 days (correct)

Why the baseline missed these. In Q09 the baseline's top 3 were s.16, s.17(1) and s.3, which only share the word "India" with the question; the real rule was 5th. In Q13 the question says "take back my permission" while the Act says "withdraw her consent". In the penalty questions the duty is in s.8 but the amount is in the Schedule at the end, and the baseline put the general rule chunks first. Where the baseline lacked the right chunk, it did not invent a rule: it either refused (Q09, Q21) or answered with a missing piece (Q13, Q25).

Re-ranking is not always better. In Q12 the cross-encoder moved the s.15 chunk ("not to impersonate") from 2nd to 7th, and in Q20 it dropped the second chunk the answer needed. In Q25 and Q03 the right chunk was in the context but the LLM still refused, so there the weak step was generation, not retrieval. In 8 of 32 questions the right chunk was not in the top 10 at all, and re-ranking cannot fix that because it only reorders those 10.

Overall (32 questions). Re-ranking improves retrieval: Hit@3 0.56 → 0.66 (18 → 21 questions) and MRR@10 0.45 → 0.55. It moved the right chunk into the top 3 in 4 questions and out of it in 1. The final answers, which I checked by hand, came out about even: 17 correct for the baseline vs. 16 after re-ranking (refused 11 vs. 13, missing a clause 2 vs. 1, wrong rule 2 vs. 2, hallucinated 0 vs. 0). So better ranking helped in the cases above, but with this LLM it did not raise overall answer accuracy.

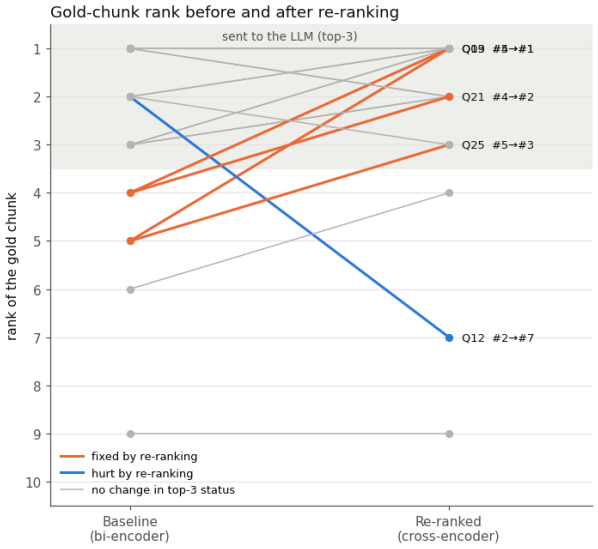


Figure 1: Rank of the correct chunk before and after re-ranking, one line per question (shaded band = top 3 sent to the LLM).

3. Latency

32 questions, 30 runs each after a warm-up. LLM time is not included (about 5 s per call, the same for both pipelines).

	GPU, RTX 3050 (ms)		CPU (ms)	
Stage	median	p95	median	p95
Pipeline 1: baseline retrieval (top 10)	7.0	8.3	15.9	19.5
Cross-encoder re-ranking (10 pairs)	27.0	28.2	329.8	375.6
Pipeline 2 total	34.1	35.9	346.2	393.4

Pipeline 2 is about 5× slower on the GPU and about 22× slower on the CPU. The baseline only encodes the short question and searches the stored vectors; the cross-encoder runs the full model once for each of the 10 pairs.

4. Scaling question

A cross-encoder score depends on the question and the chunk together, so nothing can be computed ahead of time. Running it over the whole collection means one full model pass per chunk, for every question, and the cost grows linearly with the collection size. On the GPU it took 31 ms for 10 chunks and 2.2 s for 1,000 chunks (about 2.2 ms per pair), so:

100,000 chunks ≈ 3.6 minutes per question, 1,000,000 chunks ≈ 36 minutes per question.

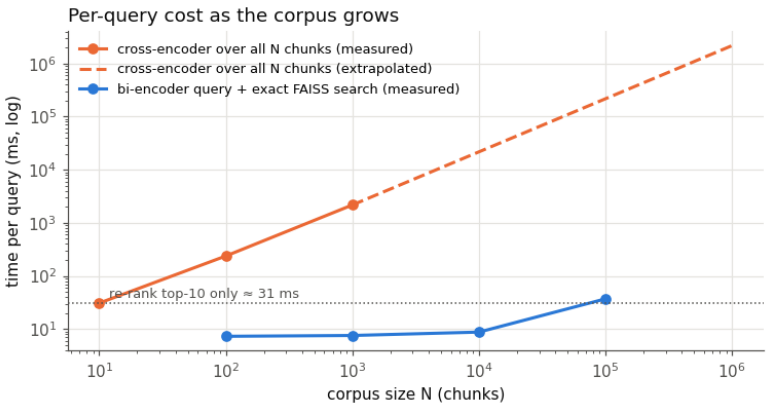


Figure 2: Per-question cost vs. collection size (log scale). Dashed line: extrapolated.

The bi-encoder has already embedded the chunks offline, so its search over 100,000 vectors still took only 37 ms. That is why the two are combined: the bi-encoder cheaply narrows the collection down to 10 candidates, and the expensive cross-encoder is used only to order those 10.